matter within a single theory capable of describing all physical phenomena.

Albert Einstein, who for more than three decades sought to combine electromagnetism and general relativity in a single theory, is rightly credited with initiating the modern search for a unified theory. For long stretches during those decades, he was the sole searcher for such a unified theory, and his passionate yet solitary quest alienated him from the mainstream physics community. During the last twenty years, though, there has been a resurgence in the quest for a unified theory; Einstein's lonely dream has become the driving force for a whole generation of physicists. But with the discoveries since Einstein's time has come a shift in focus. Even though we don't yet have a successful theory combining the strong nuclear force and the electroweak force, all three of these forces (electromagnetic, weak, strong) have been described by a single uniform language based on quantum mechanics. But general relativity, our most refined theory of the fourth force, stands outside this framework. General relativity is a classical theory: it does not incorporate any of the probabilistic concepts of quantum theory. A primary goal of the modern unification program is therefore to combine general relativity and quantum mechanics, and to describe all four forces within the same quantum mechanical framework. This has proven to be one of the most difficult problems theoretical physics has ever encountered.

Let's see why.

Quantum Jitters and Empty Space

If I had to select the single most evocative feature of quantum mechanics, I'd choose the uncertainty principle. Probabilities and wavefunctions certainly provide a radically new framework, but it's the uncertainty principle that encapsulates the break from classical physics. Remember, in the seventeenth and eighteenth centuries, scientists believed that a complete description of physical reality amounted to specifying the positions and velocities of every constituent of matter making up the cosmos. And with the advent of the field concept in the nineteenth century, and its subsequent application to the electromagnetic